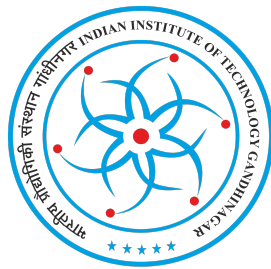


INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR (IIT Gandhinagar)



Assignment Report

M.Tech in Integrated Circuit Design and Technology (ICDT)

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Subject Name:	Analog IC Design Lab
Subject Code:	EE 653
Assignment No:	03
Submission Date:	28 January 2025
Instructor Name:	Professor Madhav Pathak

TASK 1: Designing Current Reference

The output current is defined as

$$I_{\text{out}} = 40 \times (\text{First letter of your first name} + \text{Last letter of your first name}) \text{ nA}$$

1. **Design a current reference using an ideal resistor** (refer to Fig. 1)
 - (a) Plot I_{out} with respect to V_{DD} , with V_{DD} varying from 1.44 V to 2.16 V.
2. **Design a supply-independent current reference** (refer to Fig. 2). The resistor value should not exceed 100 k Ω .
 - (a) Plot I_{out} with respect to V_{DD} , with V_{DD} varying from 1.44 V to 2.16 V.

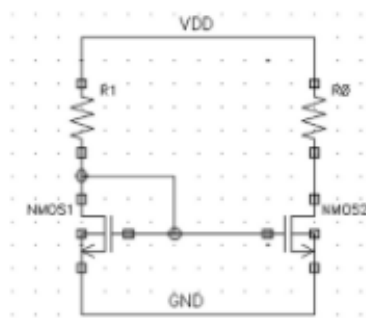


Fig. 1

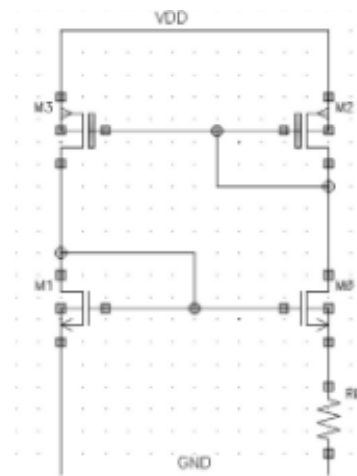


Fig. 2

Figure 1

TASK- 1.1

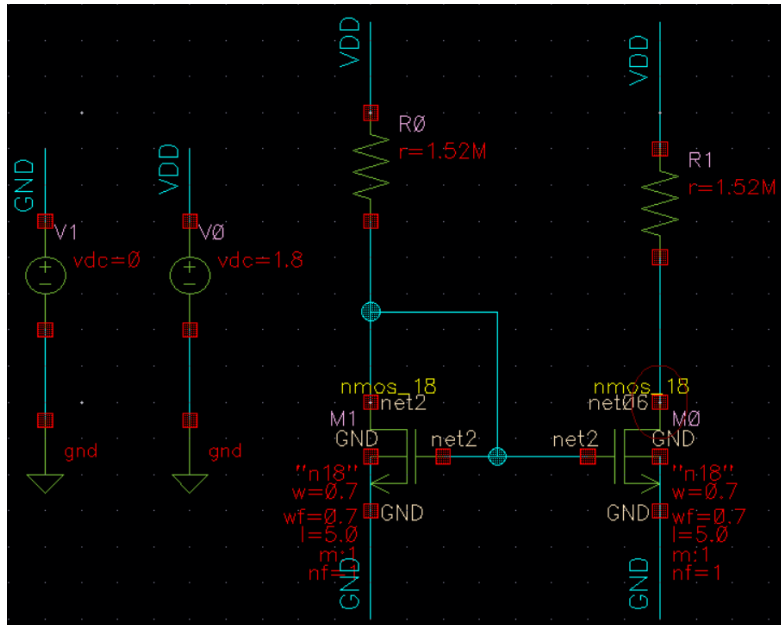


Figure 2: Schematic with component parameters

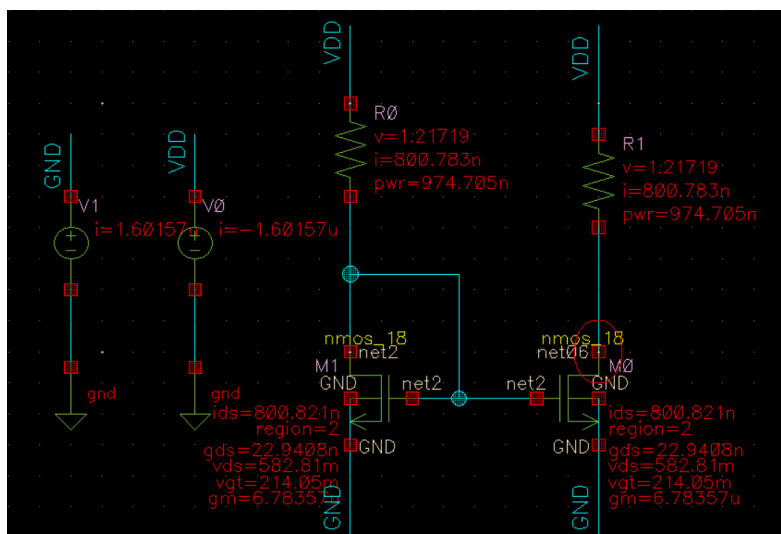


Figure 3: Schematic with DC parameters

Simulation Result

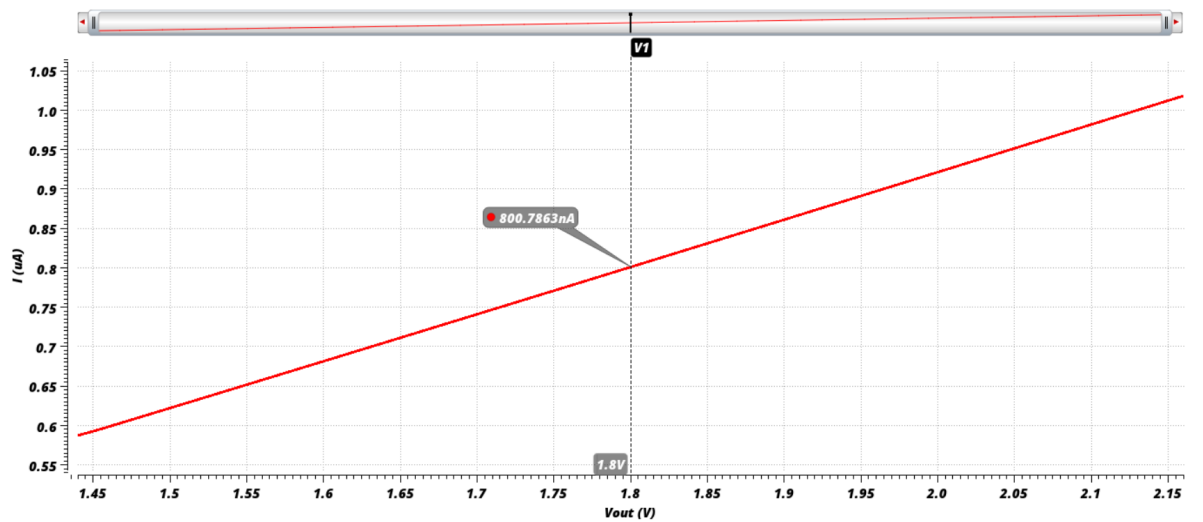


Figure 4: Varying VDD from 1.44V to 2.16V and plotting I_{out} Vs VDD

Calculations

PSRR = $+20 \log \left(\frac{v_{in}}{v_{out}} \right)$

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Mo Tu We Th Fr Sa Su

1.1

$I_{out} = 40 \times (11 + 9) = 800 \text{ nA}$

$(w/l)_1 = \frac{800 \times 10^{-9} \times 2}{300 \times 10^6 \times (0.2)^2}$

$(w/l)_2 = 0.133 \mu \Rightarrow L = 5 \mu \text{m}$
 $W = 0.718 \mu \text{m}$

$V_{ov} = V_{GS1} - V_{th1} = 0.2$
 $V_{GS1} = 0.65 \text{ V}$
 $V_{G1} = 0.65 \text{ V}$

(Theoretical)
 $W = 0.5 \mu \text{m}$
 (practical)

$V_{D2} = V_{G2} - V_{th}$
 $V_{D2} = 0.2 \text{ V}$

$V_{DD} - V_{G1} = R_1 \cdot I_{ref}$
 $R_1 = \frac{1.8 - 0.65}{800 \times 10^{-9}} = 1.4375 \text{ M}\Omega$

$R_2 = \frac{1.8 - 0.2}{800 \text{ nA}}$
 $R_2 = 2 \text{ M}\Omega$

Figure 5: Calculations of task 1.1

TASK- 1.2

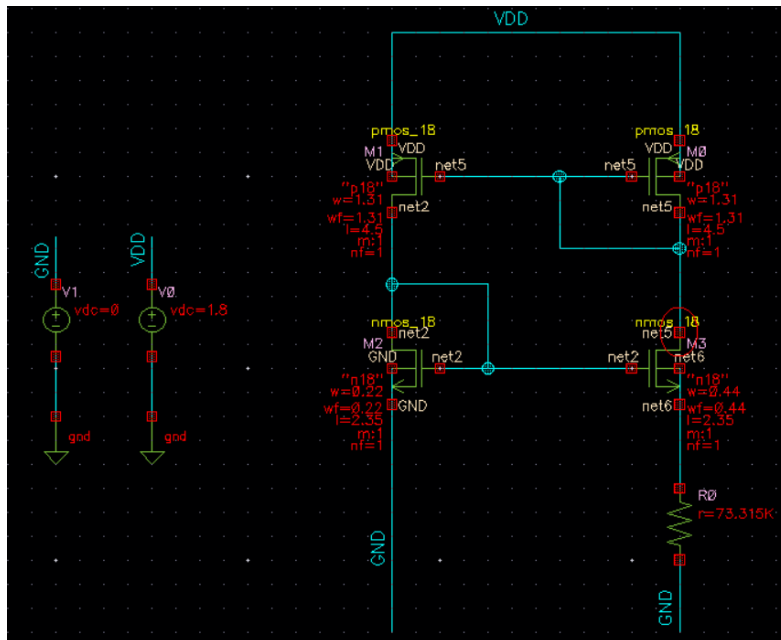


Figure 6: Schematic with component parameters

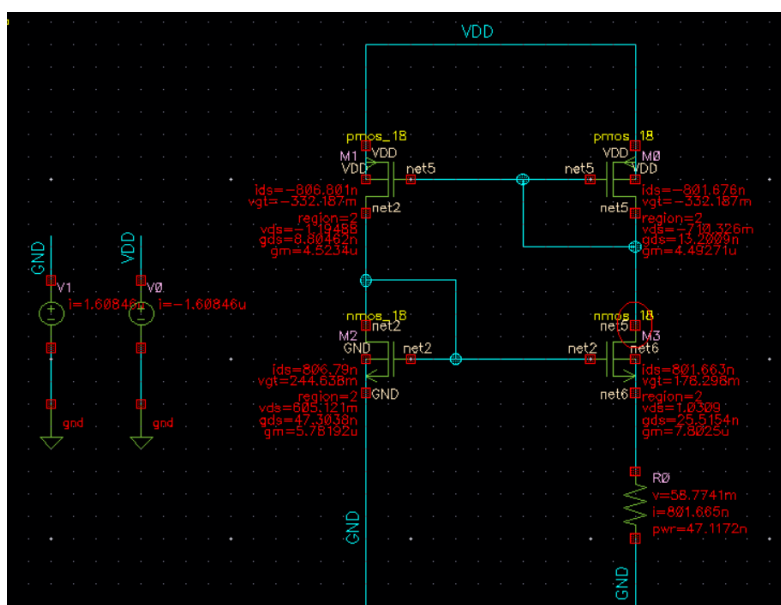


Figure 7: Schematic with DC parameters

Simulation Result

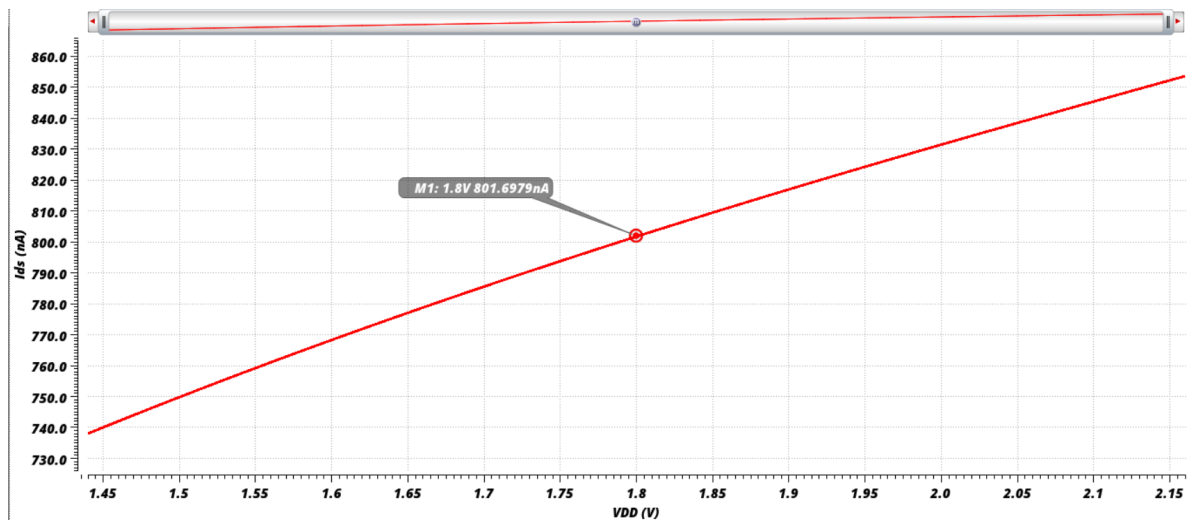
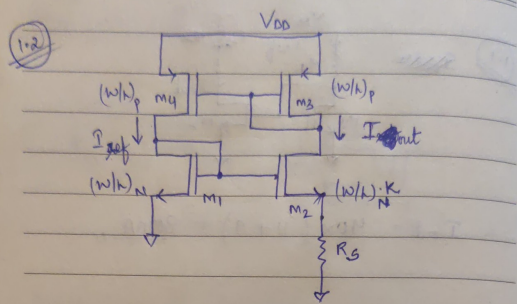


Figure 8: Varying V_{DD} from 1.44V to 2.16V and plotting I_{out} Vs V_{DD}

Calculations

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$I_{out} = \frac{1}{2} \mu_p C_{ox} \cdot (w/l)_p \cdot V_{ov}^2$
 $800 \times 10^{-9} = 65 \times 10^{-6} \times (w/l)_p \times (0.2)^2$
 $(w/l)_p = \frac{800 \times 10^{-9}}{65 \times 10^{-6} \times (0.2)^2}$
 $(w/l)_p = 0.615$
 $(w/l)_p = \frac{W}{L} \cdot \frac{1}{L_p}$
 $W = 1.31 \mu m$
 $L = 45 \mu m$
 $(w/l)_N = \frac{800 \times 10^{-9} \times 2}{300 \times 10^{-6} \times 0.2^2} = 0.133$

(a)

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$V_{ov} = V_{GS1} - V_{th1} = 0.2V$
 $V_{GS1} = 0.65V$
 $V_{ov} = V_{GS2} - V_{th2} = 0.2V$
 $V_{GS2} - (R_S \cdot I_{out}) = 0.65V$
 $R_S = \frac{V_{GS2} - 0.65V}{I_{out}}$
 $R_S = \frac{0.65V - 0.65V}{800 \times 10^{-9}A}$
 $R_S = 0 \Omega$
 $R_S = 73.315 \Omega$

(b)

Figure 9: calculations for task 1.2

TASK 2: Designing Sub-1-V BGR using OTA

1. Using the Temperature Coefficient of CTAT and PTAT obtained after simulation in Assignment-2, design a Sub-1-V BGR (refer to Fig. 3) with an output BGR voltage less than 1 V. The design should meet the following specifications, and the simulated measurements should be reported/plotted. (The output voltage of the BGR should be according to the table provided.)
 - (a) Temperature regulation: 10 mV (for temperature variation from -40°C to 125°C)
 - (b) Supply voltage regulation: 10 mV (for supply voltage variation of ± 0.1 V)
 - (c) Power budget: $50\text{ }\mu\text{W}$
 - (d) Supply voltage: 1.8 V
 - (e) PSRR at 1 MHz > 25 dB
2. Include the derivation (digitally written or scanned) in the report and justify the choice of design parameters such as resistor values, multiplier, etc.
3. Design the circuit for the **tt** (typical–typical) process corner. Simulate and analyze the circuit for the remaining four corners (**ff**, **ss**, **fs**, and **sf**) and include the temperature regulation and supply voltage regulation for all corners in a tabular format in the report. (Corner simulations are mandatory; however, it is acceptable if the specifications are not met for all corners.)

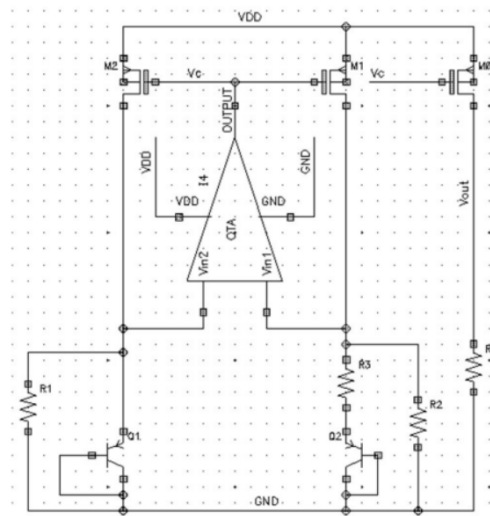


Fig. 3

Figure 10

Schematic:

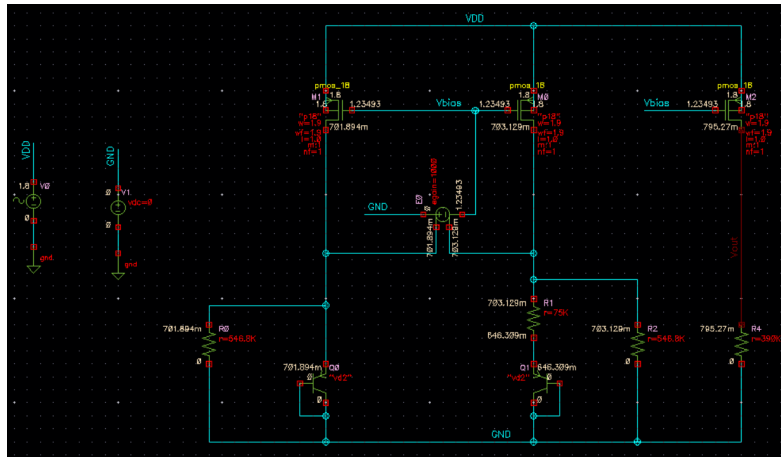


Figure 11: Schematic with component parameters

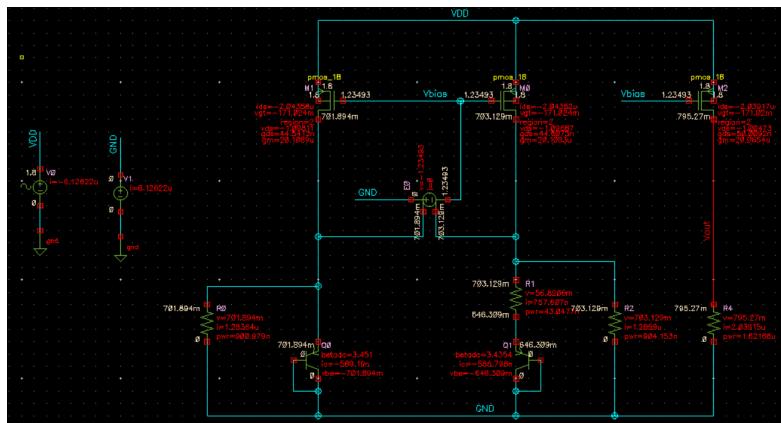


Figure 12: Schematic with DC parameters

Simulation Result

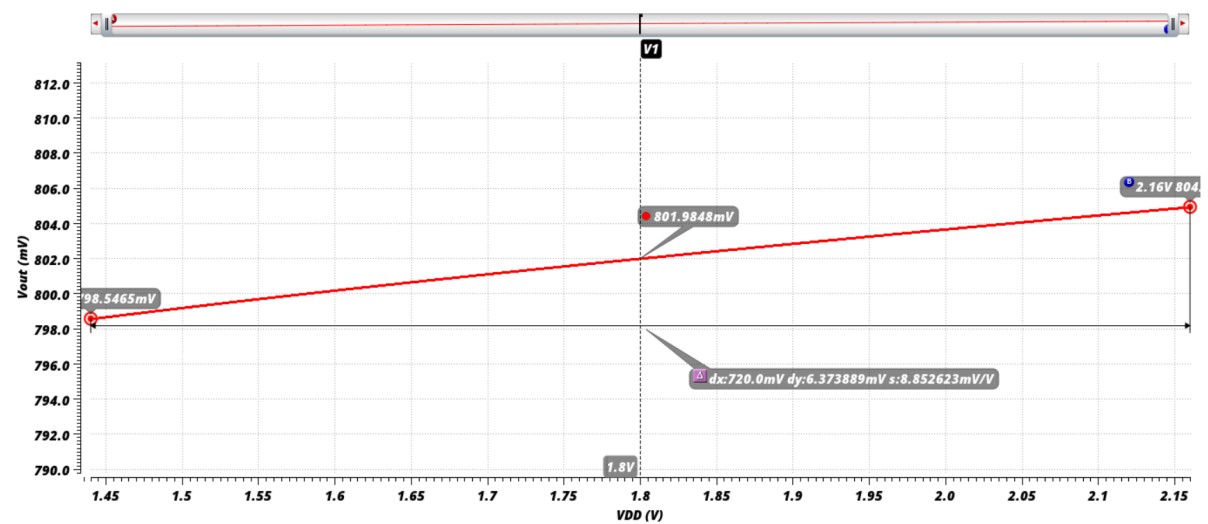


Figure 13: Voltage Regulation

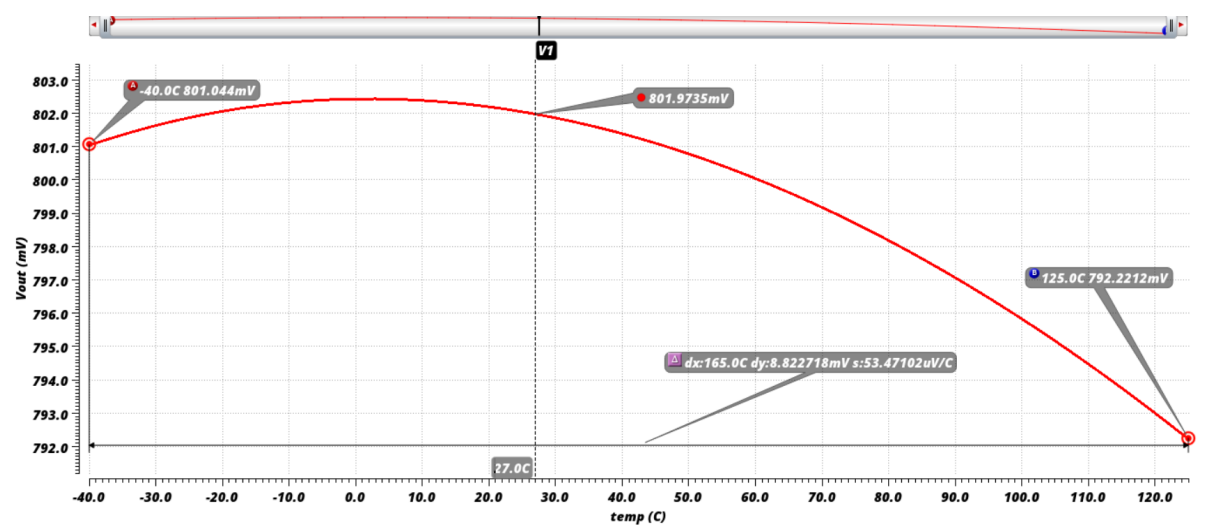


Figure 14: Temperature Regulation

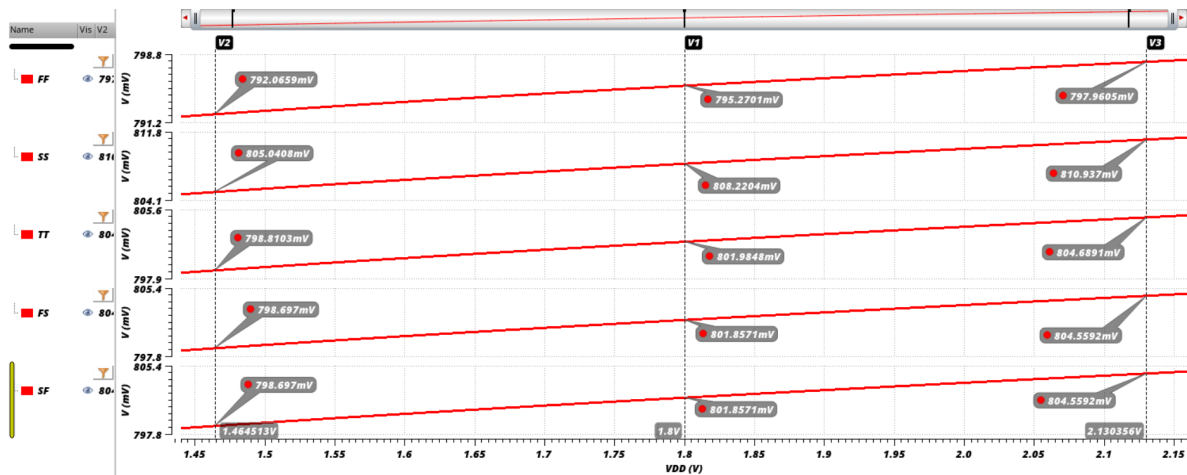


Figure 15: Voltage Regulation at different Process Corners

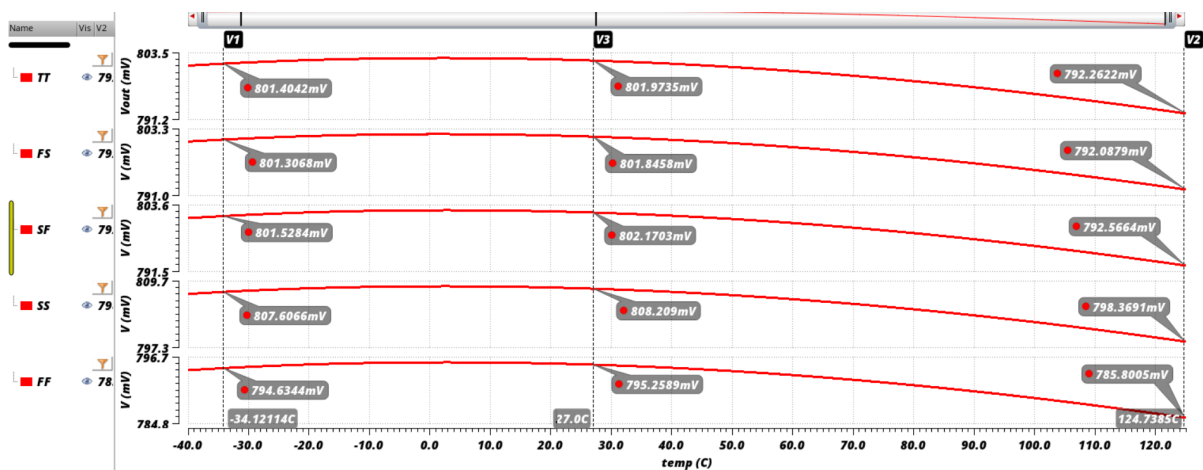


Figure 16: Temperature Regulation at different Process Corners

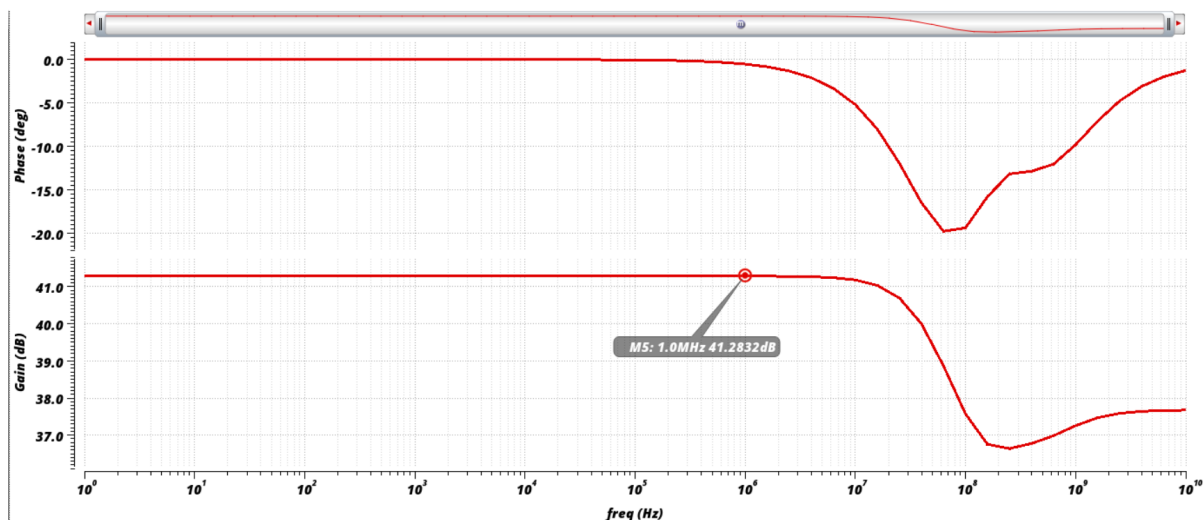


Figure 17: PSRR

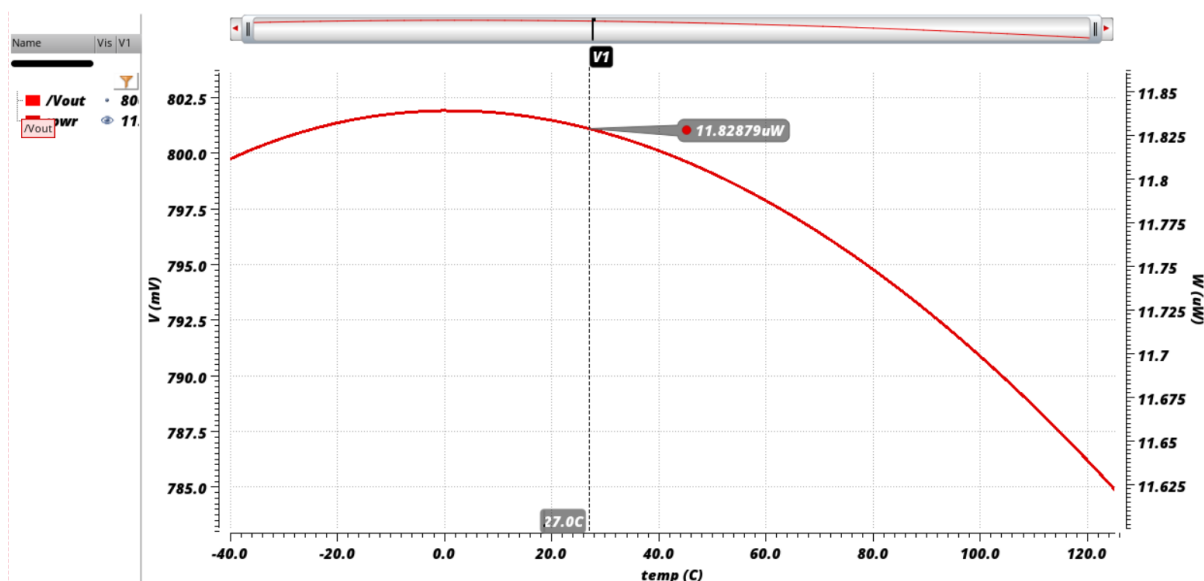
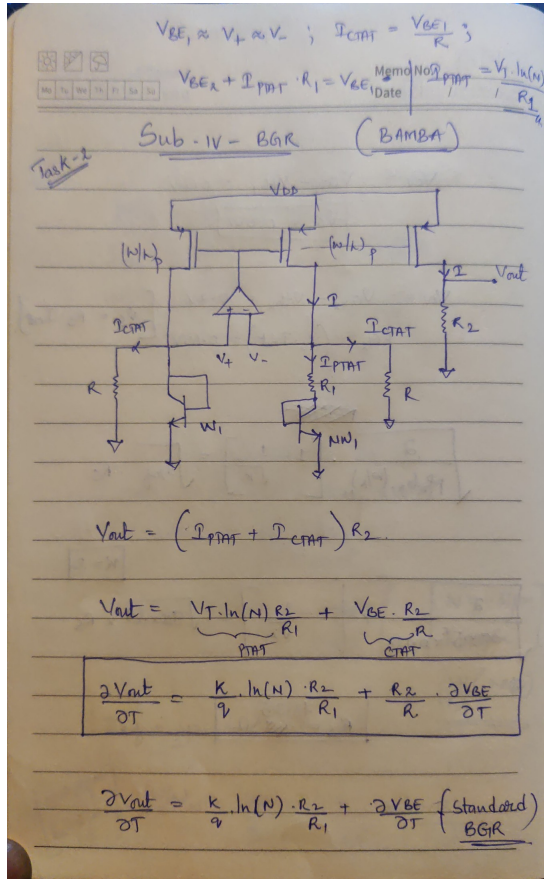


Figure 18: Power Vs Temperature

Calculations



(a)

$R_2 = \frac{V_{out}}{I_{out}} = \frac{800 \times 10^{-3}}{2 \times 10^{-6}} = 400 \text{ k}\Omega$
 $I_b = \frac{1}{2} \cdot \mu_p \cdot C_{ox} \cdot \left(\frac{W}{L} \right)_p \cdot V_{ov}^2$
 $2 \times 10^{-6} = \frac{1}{2} \times 65 \times 10^{-6} \times \left(\frac{W}{L} \right)_p \cdot (0.2)^2$
 $\boxed{\left(\frac{W}{L} \right)_p = 1.538}$
 $V_{out} = V_T \cdot \ln(N) \cdot \frac{R_2}{R_1} + V_{BE} \cdot \frac{R_2}{R}$
 $\frac{\partial V_{out}}{\partial T} = \frac{K}{q} \cdot \ln(8) \cdot \frac{400 \text{ k}\Omega}{R_1} + \frac{\partial V_{BE}}{\partial T} = 0$
 $R_1 = \frac{86 \times 10^{-6} \times \ln(8) \times 400 \text{ k}\Omega}{+1.35 \times 10^{-3}}$
 $\boxed{R_1 = 52.987 \text{ k}\Omega}$
 $V_{out} = V_T \cdot \ln(8) \cdot \frac{R_2}{R_1} + V_{BE} \cdot \frac{R_2}{R} = 0.8$
 $0.8 - 0.026 \times \ln(8) \times \frac{400}{52.987} = 0.7 \times \frac{400}{R}$
 $R = \frac{0.7 \times 400}{0.391} \Rightarrow \boxed{R = 716.11 \text{ k}\Omega}$

(b)

Figure 19: calculations for task 2

Table 1: Calculated and Simulated Values

Device	$(W/L)_{\text{calculated}}$	$(W/L)_{\text{used}}$	V_{ov} (mV)
M0	1.538	1.9	203.278
M1	1.538	1.9	203.278
M2	1.538	1.9	203.278

Table 2: Calculated and Used Resistance Values

Resistor	Calculated Value	Used Value
R_1	52.987k Ω	75k Ω
R_4	400k Ω	366k Ω
R_2/R_0	716.11k Ω	546.8k Ω

Table 3: Performance Summary of Sub-1-V BGR

Parameter	Value
Bandgap reference voltage	800 mV
Line regulation	6.3 mV
Temperature variation	8.8 mV
PSRR at 1 MHz	41.28 dB
Power consumption	11.82 μ W

Conclusion

The designed sub-1-V bandgap reference generates an output voltage of 800 mV. The circuit exhibits good line regulation of 6.3 mV and temperature variation of 8.8 mV, demonstrating stable operation across supply voltage and temperature changes. A PSRR of 41.28 dB at 1 MHz is achieved, ensuring effective suppression of supply-induced noise. The total power consumption is 11.82 μ W.

Overall, the proposed bandgap reference shows minimal sensitivity to supply and temperature variations compared to the Task-1 bandgap reference circuits. Owing to its high stability, low power consumption, and adequate PSRR, the circuit is well suited for use as a reference voltage source.